

HeartGuard Chapter 3 Design Diagrams

HEARTGUARD: Preventive Insights for Heart Failure Readmission Forecasting using Machine Learning

White-background figures prepared for Chapter 3: Analysis and Design

Figure 3.1: System Architecture

Layered architecture for HeartGuard

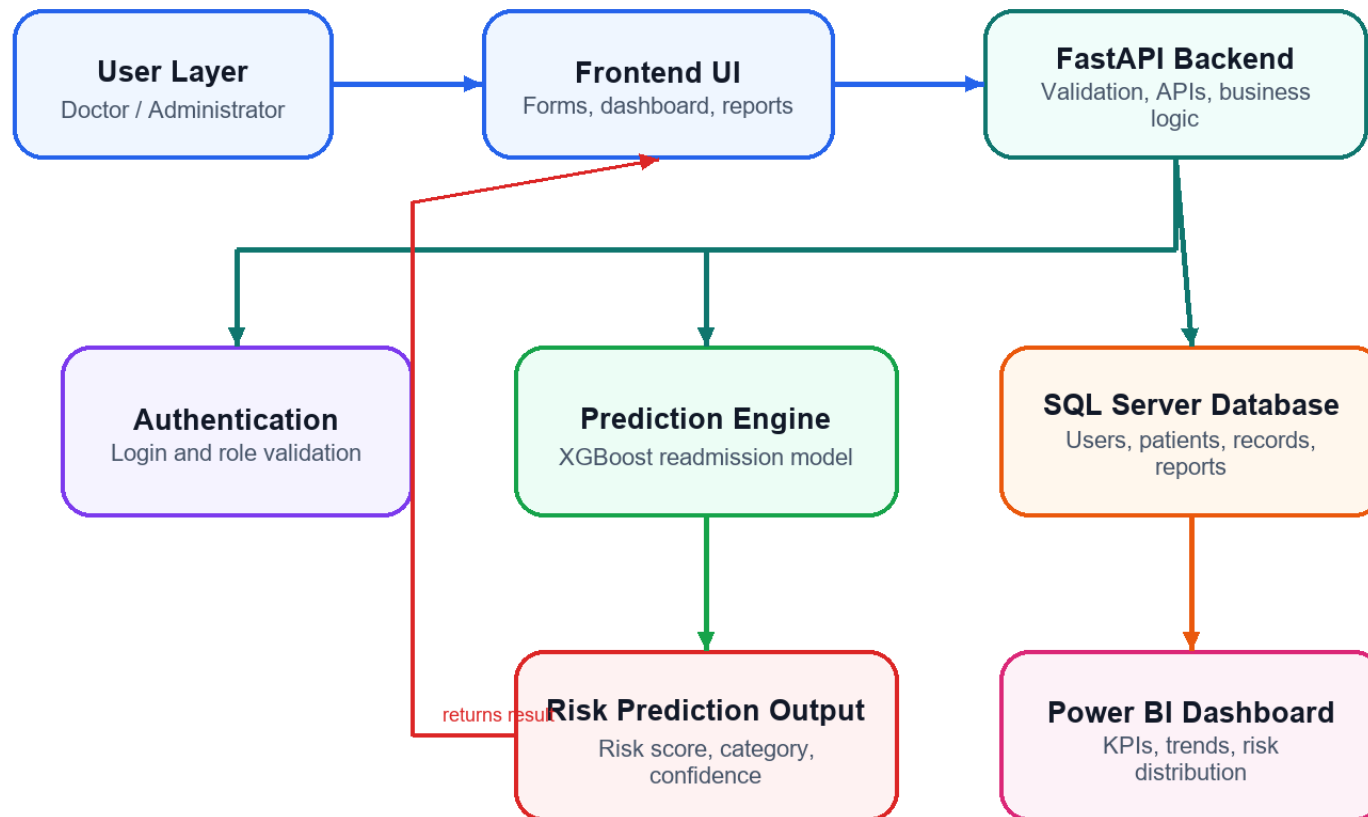


Figure 3.1: System Architecture of HeartGuard

Figure 3.2: Use Case Diagram

Actors and major system interactions

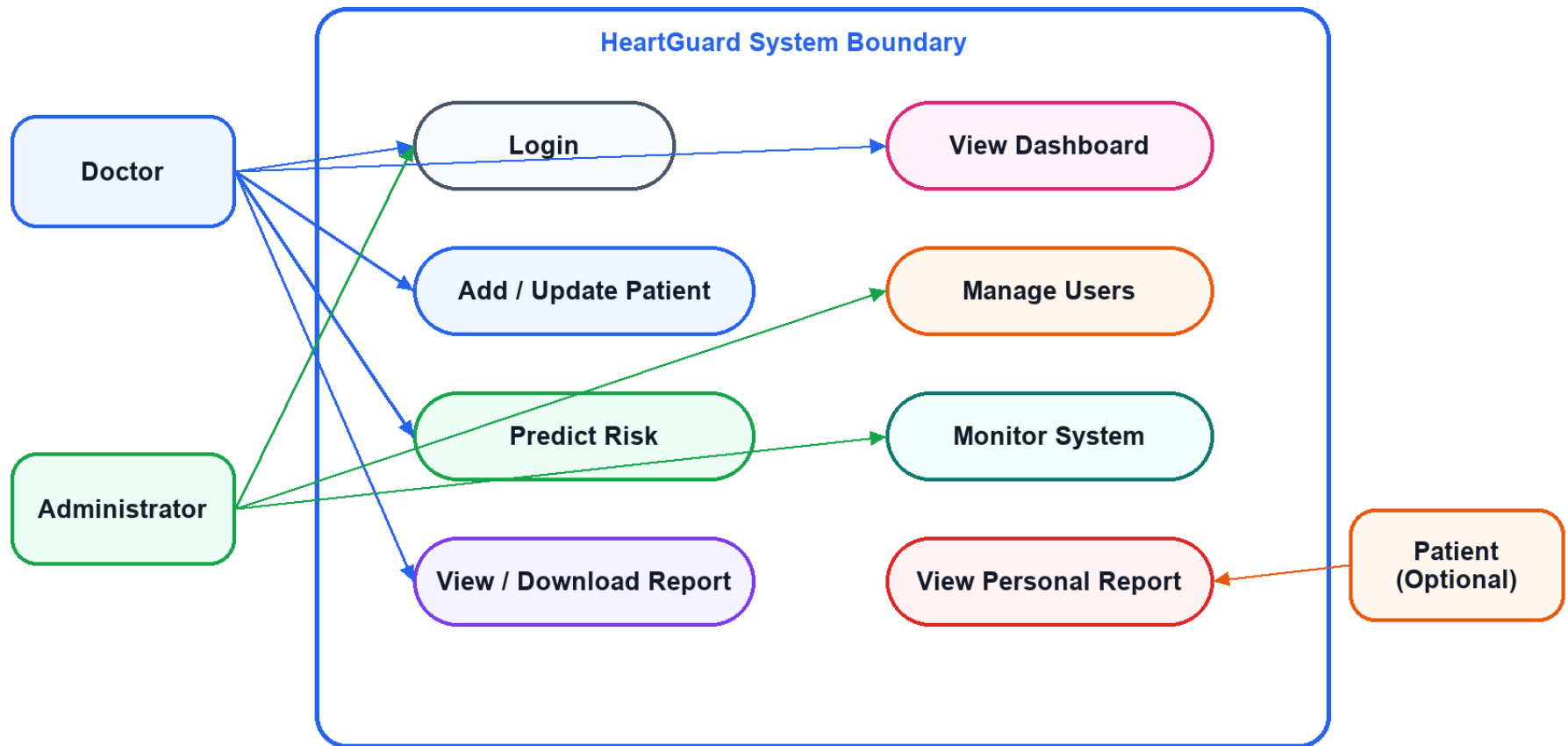


Figure 3.2: Use Case Diagram of HeartGuard

Figure 3.3: Data Flow Diagram

Movement of data from input to clinical insight

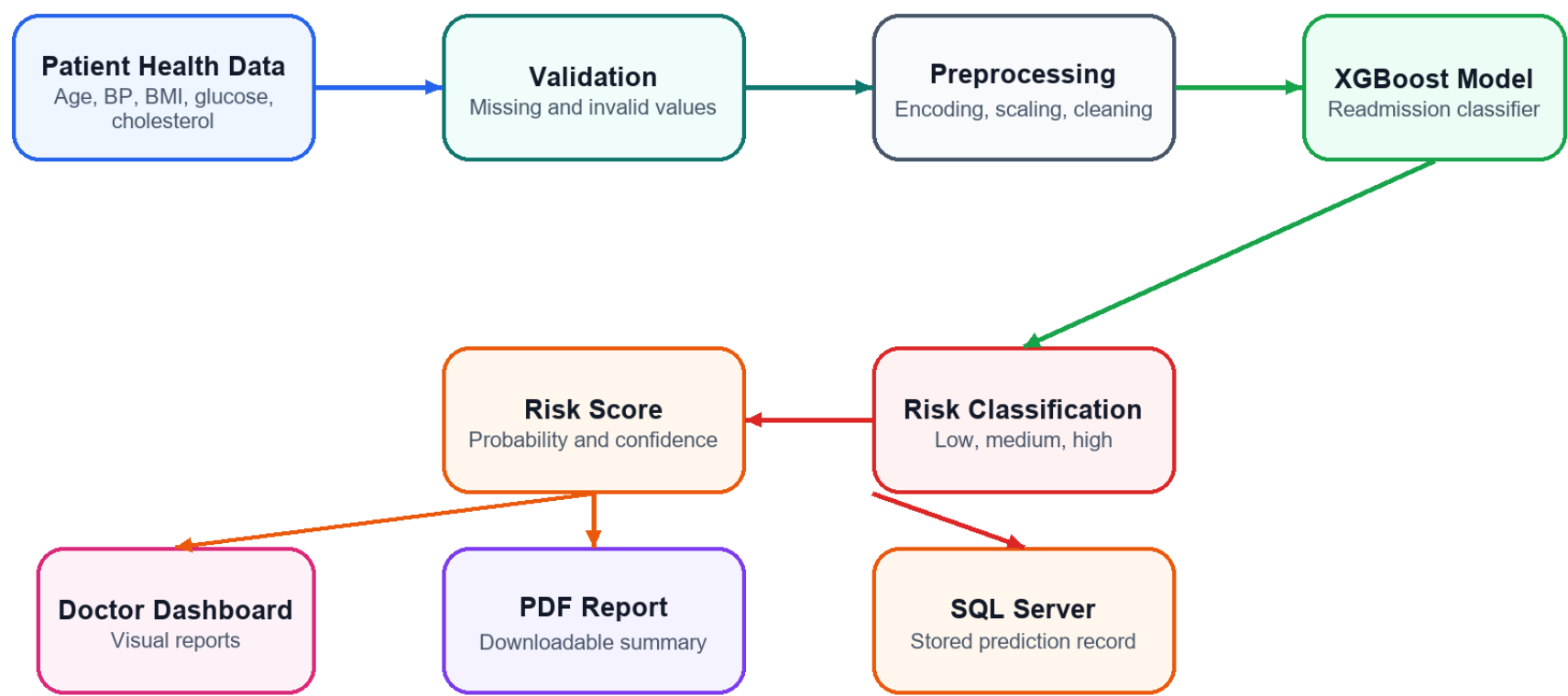


Figure 3.3: Data Flow Diagram of HeartGuard

Figure 3.4: Entity Relationship Diagram

Database entities used by HeartGuard

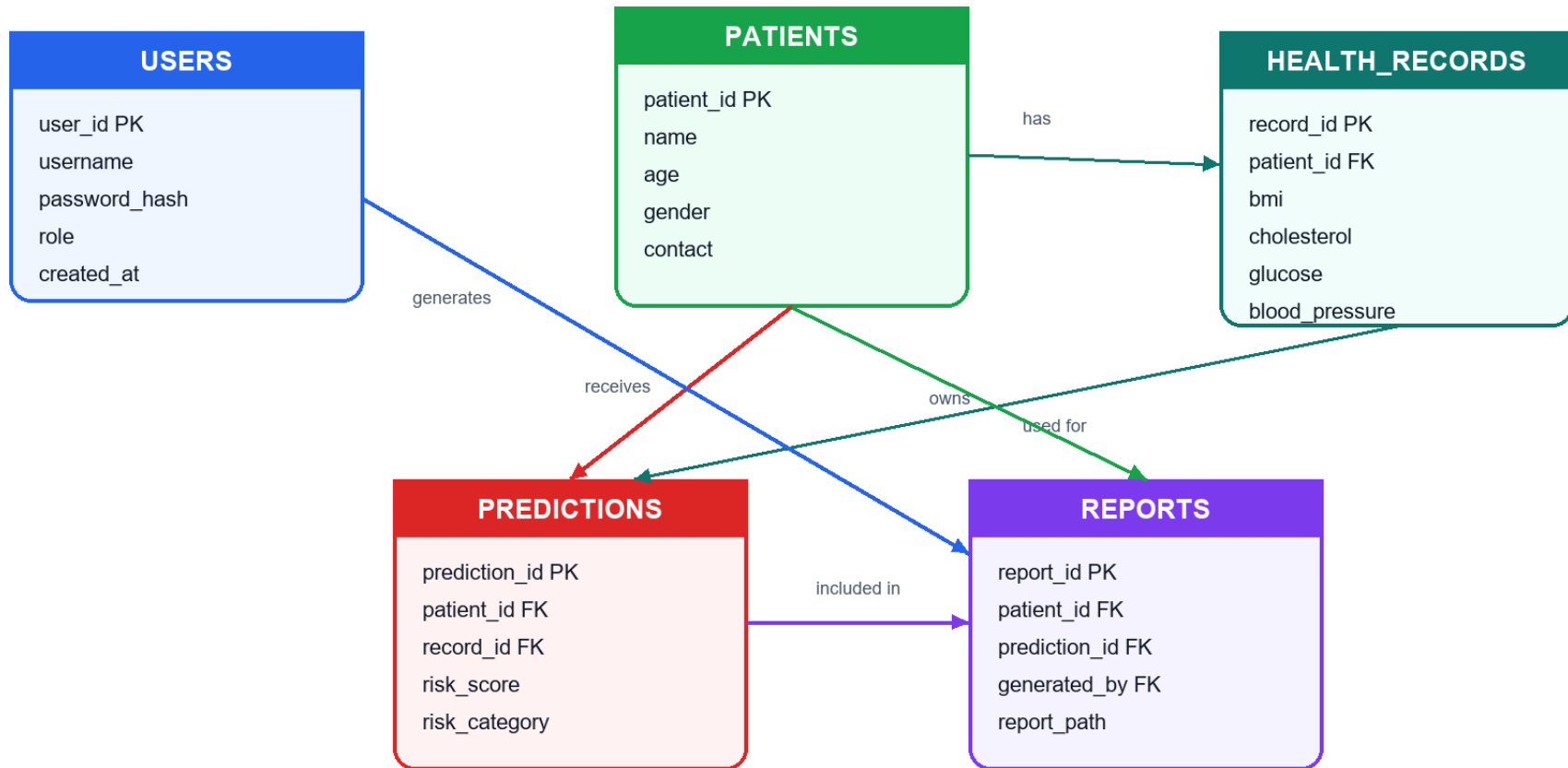


Figure 3.4: Entity Relationship Diagram of HeartGuard Database

Figure 3.5: Prediction Workflow

Operational workflow for generating a readmission forecast

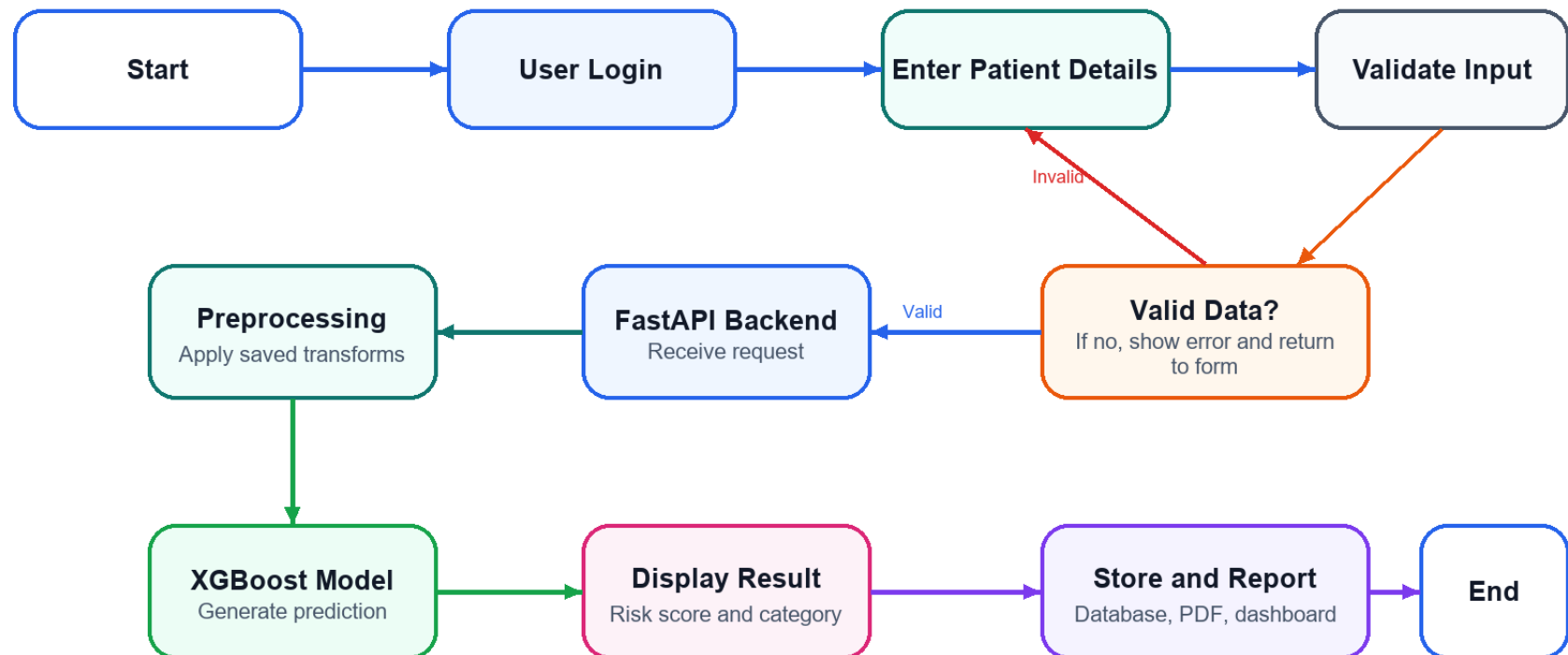


Figure 3.5: HeartGuard Prediction Workflow